EECS 16A Foundations of Signals, Systems, & Information Processing
Fall 2025 Homework 4

This homework is due Friday, September 26, 2025, at 23:59.

Submission Format

Your homework submission should consist of a single PDF file that contains all of your answers (any handwritten answers should be scanned) as well as your IPython notebook saved as a PDF.

If you do not attach a PDF "printout" of your IPython notebook, you will not receive credit for problems that involve coding. Make sure that your results and your plots are visible. Assign the IPython printout to the correct problem(s) on Gradescope.

1. Vector Statistics

Let $\mathbf{x}$ be vector in $\mathbb{R}^n$. We define the average of a vector, $avg(\mathbf{x})$, to be

$$avg(\mathbf{x}) = \frac{x_1 + x_2 + \ldots x_n}{n} = \frac{\langle \mathbf{x}, \mathbf{1} \rangle}{n}$$

Similarly, we define the root mean square value of a vector, $rms(\mathbf{x})$, to be:

$$rms(\mathbf{x}) = \sqrt{\frac{x_1^2 + x_2^2 + \ldots x_n^2}{n}} = \frac{\|\mathbf{x}\|}{\sqrt{n}}$$

Finally, we define the standard deviation of a vector, $std(\mathbf{x})$, to be:

$$std(\mathbf{x}) = \sqrt{\frac{(x_1 - avg(\mathbf{x}))^2 + (x_2 - avg(\mathbf{x}))^2 + \ldots (x_n - avg(\mathbf{x}))^2}{n}} = \frac{\|\mathbf{x} - avg(\mathbf{x})\mathbf{1}\|}{\sqrt{n}} = rms(\mathbf{x} - avg(\mathbf{x})\mathbf{1})$$

Recall that $\mathbf{1}$ is a vector of 1s. $\mathbf{x} - avg(\mathbf{x})\mathbf{1}$ is also called the demeaned vector of $\mathbf{x}$ and denoted by $\tilde{\mathbf{x}}$. More information about vector statistics is found in VMLS Chapter 3.1, 3.3, and 3.4.

(a) Calculate $avg(\mathbf{x})$, $rms(\mathbf{x})$ and $std(\mathbf{x})$, where $\mathbf{x} = \begin{bmatrix} 4 \\ 0 \\ 2 \\ 2 \end{bmatrix}$

Solution:

$$avg(\mathbf{x}) = \frac{4+0+2+2}{4} = 2$$

$$rms(\mathbf{x}) = \sqrt{\frac{4^2+0+2^2+2^2}{4}} = \sqrt{6}$$

$$
\begin{aligned}
std(\mathbf{x}) &= \sqrt{\frac{(4-2)^2+(0-2)^2+(2-2)^2+(2-2)^2}{4}} \\
&= \sqrt{\frac{(2)^2+(-2)^2+(0)^2+(0)^2}{4}} \\
&= \sqrt{2}
\end{aligned}
$$

For the following parts, let $\mathbf{x}$ and $\mathbf{y}$ be two vectors in $\mathbb{R}^n$.

(b) Show that $avg(\mathbf{x}+\mathbf{y}) = avg(\mathbf{x}) + avg(\mathbf{y})$.

Solution: We have that:

$$
\begin{aligned}
avg(\mathbf{x}+\mathbf{y}) &= \frac{\langle \mathbf{x}+\mathbf{y}, \mathbf{1} \rangle}{n} \\
&= \frac{\langle \mathbf{x}, \mathbf{1} \rangle + \langle \mathbf{y}, \mathbf{1} \rangle}{n} \\
&= \frac{\langle \mathbf{x}, \mathbf{1} \rangle}{n} + \frac{\langle \mathbf{y}, \mathbf{1} \rangle}{n} \\
&= avg(\mathbf{x}) + avg(\mathbf{y})
\end{aligned}
$$

(c) Show that $avg(\mathbf{x})^2 + std(\mathbf{x})^2 = rms(\mathbf{x})^2$.

Hint: Start with $std(\mathbf{x})^2$ and expand the equation using inner products.

Solution: We have that:

$$
\begin{aligned}
std(\mathbf{x})^2 &= \frac{\|\mathbf{x} - avg(\mathbf{x})\mathbf{1}\|^2}{n} \\
&= \frac{\langle \mathbf{x} - avg(\mathbf{x})\mathbf{1}, \mathbf{x} - avg(\mathbf{x})\mathbf{1} \rangle}{n} \\
&= \frac{\langle \mathbf{x}, \mathbf{x} \rangle - 2\langle \mathbf{x}, avg(\mathbf{x})\mathbf{1} \rangle + \langle avg(\mathbf{x})\mathbf{1}, avg(\mathbf{x})\mathbf{1} \rangle}{n} \\
&= \frac{\|\mathbf{x}\|^2 - 2avg(\mathbf{x})\langle \mathbf{x}, \mathbf{1} \rangle + n \cdot avg(\mathbf{x})^2}{n} \\
&= \frac{\|\mathbf{x}\|^2 - 2n \cdot avg(\mathbf{x})^2 + n \cdot avg(\mathbf{x})^2}{n} \\
&= \frac{\|\mathbf{x}\|^2 - n \cdot avg(\mathbf{x})^2}{n} \\
&= \frac{\|\mathbf{x}\|^2}{n} - avg(\mathbf{x})^2 \\
&= rms^2(\mathbf{x}) - avg(\mathbf{x})^2
\end{aligned}
$$

Therefore we get that $avg(\mathbf{x})^2 + std(\mathbf{x})^2 = rms(\mathbf{x})^2$.

(d) If $\mathbf{x}$ and $\mathbf{y}$ are orthogonal, show that $rms(\mathbf{x}+\mathbf{y}) = \sqrt{rms(\mathbf{x})^2 + rms(\mathbf{y})^2}$.
Hint: We have that $\|\mathbf{x}+\mathbf{y}\|^2 = \|\mathbf{x}\|^2 + 2\langle \mathbf{x}, \mathbf{y}\rangle + \|\mathbf{y}\|^2$.
Solution: By expanding out $rms(\mathbf{x}+\mathbf{y})^2$ and using the hint, we get:

$$\begin{aligned} rms(\mathbf{x}+\mathbf{y})^2 &= \frac{\|\mathbf{x}+\mathbf{y}\|^2}{n} \\ &= \frac{\|\mathbf{x}\|^2 + 2\langle \mathbf{x}, \mathbf{y}\rangle + \|\mathbf{y}\|^2}{n} \\ &= \frac{\|\mathbf{x}\|^2 + \|\mathbf{y}\|^2}{n} \\ &= \frac{\|\mathbf{x}\|^2}{n} + \frac{\|\mathbf{y}\|^2}{n} \\ &= rms(\mathbf{x})^2 + rms(\mathbf{y})^2 \end{aligned}$$

Taking the square root of both sides gives us $rms(\mathbf{x}+\mathbf{y}) = \sqrt{rms(\mathbf{x})^2 + rms(\mathbf{y})^2}$.

(e) If $\tilde{\mathbf{x}}$ and $\tilde{\mathbf{y}}$ (the demeaned vectors of $\mathbf{x}$ and $\mathbf{y}$) are orthogonal, show that $std(\mathbf{x}+\mathbf{y}) = \sqrt{std(\mathbf{x})^2 + std(\mathbf{y})^2}$.
Hint: Express $std(\mathbf{x}+\mathbf{y})$ *in terms of rms and apply the previous part.*
Solution: Since $std(\mathbf{x}) = rms(\mathbf{x} - avg(\mathbf{x})\mathbf{1})$, we can use the previous part to get:

$$\begin{aligned} std(\mathbf{x}+\mathbf{y})^2 &= rms(\mathbf{x}+\mathbf{y} - avg(\mathbf{x}+\mathbf{y})\mathbf{1})^2 \\ &= rms(\mathbf{x}+\mathbf{y} - avg(\mathbf{x})\mathbf{1} - avg(\mathbf{y})\mathbf{1})^2 \\ &= rms((\mathbf{x} - avg(\mathbf{x})\mathbf{1}) + (\mathbf{y} - avg(\mathbf{y})\mathbf{1}))^2 \\ &= rms(\mathbf{x} - avg(\mathbf{x})\mathbf{1})^2 + rms(\mathbf{y} - avg(\mathbf{y})\mathbf{1})^2 \\ &= std(\mathbf{x})^2 + std(\mathbf{y})^2 \end{aligned}$$

Taking the square root of both sides gives us $std(\mathbf{x}+\mathbf{y}) = \sqrt{std(\mathbf{x})^2 + std(\mathbf{y})^2}$.

(f) If $\mathbf{x}$ and $\mathbf{y}$ are linearly dependent, facing the same direction, and nonzero, show that $std(\mathbf{x}+\mathbf{y}) = std(\mathbf{x}) + std(\mathbf{y})$.
Solution:
If $\mathbf{x}$ and $\mathbf{y}$ are linearly dependent, then one vector can be written as scalar multiple of the other. Therefore suppose that $c\mathbf{x} = \mathbf{y}$ where $c \in \mathbb{R}$. Since they are facing the same direction, we have that c is positive.

Then we have that $\mathbf{x}+\mathbf{y}=\mathbf{x}+c\mathbf{x}=(c+1)\mathbf{x}$. Applying this to the definition, we get:

$$\begin{aligned}
std(\mathbf{x}+\mathbf{y}) &= std((c+1)\mathbf{x}) \\
&= \frac{\|(c+1)\mathbf{x}-avg((c+1)\mathbf{x})\mathbf{1}\|}{\sqrt{n}} \\
&= \frac{\|(c+1)(\mathbf{x}-avg(\mathbf{x})\mathbf{1})\|}{\sqrt{n}} \\
&= |c+1|\frac{\|(\mathbf{x}-avg(\mathbf{x})\mathbf{1}\|}{\sqrt{n}} \\
&= (c+1)std(\mathbf{x}) \\
&= std(\mathbf{x})+cstd(\mathbf{x}) \\
&= std(\mathbf{x})+std(c\mathbf{x}) \\
&= std(\mathbf{x})+std(\mathbf{y})
\end{aligned}$$

Therefore we get $std(\mathbf{x}+\mathbf{y})=std(\mathbf{x})+std(\mathbf{y})$ as desired.
For the general case of linear dependence, if $c<0$, then we get that $|c+1|std(\mathbf{x})$ either equals $(|c|-1|)std(\mathbf{x})$ or $(1-|c|)std(\mathbf{x})$ depending on if $|c|$ is greater or smaller than 1. This is equivalent to $|(1-|c|)std(\mathbf{x})|$, which by distributing terms gives us $|std(\mathbf{x})-|c|std(\mathbf{x})|=|std(\mathbf{x})-std(\mathbf{y})|$.
In this case, we get that $std(\mathbf{x}+\mathbf{y})=|std(\mathbf{x})-std(\mathbf{y})|$.

2. Orthonormal Norms (VMLS 5.4)

The vector $\mathbf{x}$ can be written as a linear combination of orthonormal vectors $\mathbf{v}_1, \ldots, \mathbf{v}_n$ as follows:

$$\mathbf{x} = a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \ldots + a_n\mathbf{v}_n$$

Express $\|\mathbf{x}\|$ in terms of $a_1, \ldots, a_n$.

Solution:

$$\begin{aligned}
\|\mathbf{x}\|^2 &= \|a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \ldots + a_n\mathbf{v}_n\|^2 \\
&= \langle a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \ldots + a_n\mathbf{v}_n, a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \ldots + a_n\mathbf{v}_n \rangle \\
&= \langle a_1\mathbf{v}_1, a_1\mathbf{v}_1 \rangle + \langle a_2\mathbf{v}_2, a_2\mathbf{v}_2 \rangle + \ldots + \langle a_n\mathbf{v}_n, a_n\mathbf{v}_n \rangle + \langle a_1\mathbf{v}_1, a_2\mathbf{v}_2 \rangle + \ldots + \langle a_1\mathbf{v}_1, a_n\mathbf{v}_n \rangle + \ldots \\
& \quad \mathbf{v}_1, \ldots, \mathbf{v}_n \text{ are orthogonal to each other, so their inner products are } 0. \\
&= \langle a_1\mathbf{v}_1, a_1\mathbf{v}_1 \rangle + \langle a_2\mathbf{v}_2, a_2\mathbf{v}_2 \rangle + \ldots + \langle a_n\mathbf{v}_n, a_n\mathbf{v}_n \rangle + 0 + \ldots + 0 + \ldots \\
&= \langle a_1\mathbf{v}_1, a_1\mathbf{v}_1 \rangle + \langle a_2\mathbf{v}_2, a_2\mathbf{v}_2 \rangle + \ldots + \langle a_n\mathbf{v}_n, a_n\mathbf{v}_n \rangle \\
&= \|a_1\mathbf{v}_1\|^2 + \|a_2\mathbf{v}_2\|^2 + \ldots + \|a_n\mathbf{v}_n\|^2 \\
&= a_1^2\|\mathbf{v}_1\|^2 + a_2^2\|\mathbf{v}_2\|^2 + \ldots + a_n^2\|\mathbf{v}_n\|^2 \\
&= a_1^2 + a_2^2 + \ldots + a_n^2 \\
&= \sum_{i=1}^{n} a_i^2 \\
\|\mathbf{x}\| &= \sqrt{\sum_{i=1}^{n} a_i^2}
\end{aligned}$$

From line 3 to line 4, we used the fact that $\mathbf{v}_1, \ldots, \mathbf{v}_n$ are orthogonal to each other, so their inner products are 0.

3. Orthogonal Constructions

Show that there exists a set of vectors $V = \{\mathbf{v}_1, \mathbf{v}_2, ..., \mathbf{v}_{2n}\}$ such that V spans $\mathbb{R}^n$ and $\langle \mathbf{v}_i, \mathbf{v}_j \rangle \leq 0$ for all $i \neq j$.

Hint: Is there a simple set of n vectors that satisfies our requirement? How can we expand this to $2n$ *vectors?*

Solution: Let $\mathbf{v}_1 ... \mathbf{v}_n$ form an orthogonal basis of $\mathbb{R}^n$. Using this, define $\mathbf{v}_{n+1} = -\mathbf{v}_1, \mathbf{v}_{n+2} = -\mathbf{v}_2, ... \mathbf{v}_{2n} = -\mathbf{v}_n$. Note that $\mathbf{v}_{n+1}, \mathbf{v}_{n+2} ... \mathbf{v}_{2n}$ also form an orthogonal basis for $\mathbb{R}^n$

Then consider $\langle \mathbf{v}_i, \mathbf{v}_j \rangle$ where $i \neq j$. Assume that $i < j$ since switching the vectors in the inner product does not change the value of the inner product.

We can break this down into three cases:

(a) If $i, j \leq n$ or $i, j > n$

In this case, we have $\mathbf{v}_i, \mathbf{v}_j$ are part of the orthogonal basis $\mathbf{v}_1 ... \mathbf{v}_n$ or the orthogonal basis $\mathbf{v}_{n+1}, ... \mathbf{v}_{2n}$. Since the two vectors are not equal, by the definition of an orthogonal basis, we have that their inner product must be 0, so $\langle \mathbf{v}_i, \mathbf{v}_j \rangle = 0$.

(b) If $j > n, i \leq n$ and $i + n \neq j$.

This means that $\mathbf{v}_j \neq \mathbf{v}_{i+n} = -\mathbf{v}_i$. Since $\mathbf{v}_{i+n}$ and $\mathbf{v}_j$ are distinct vectors in the orthogonal basis $\mathbf{v}_{n+1}, \mathbf{v}_{n+2} ... \mathbf{v}_{2n}$, we have that $\langle \mathbf{v}_{i+n}, \mathbf{v}_j \rangle = 0$. Therefore $\langle \mathbf{v}_i, \mathbf{v}_j \rangle = \langle -\mathbf{v}_{i+n}, \mathbf{v}_j \rangle = -\langle \mathbf{v}_{i+n}, \mathbf{v}_j \rangle = 0$.

(c) If $j > n, i \leq n$ and $i + n = j$.

This means that our two vectors are linearly dependent, as we have $\mathbf{v}_i$ and $\mathbf{v}_j = \mathbf{v}_{i+n} = -\mathbf{v}_i$. It follows that their inner product is:

$$\langle \mathbf{v}_i, -\mathbf{v}_i \rangle = -\|\mathbf{v}_i\|^2 \leq 0$$

4. Matching Frequencies

(a) Let $\omega \in \mathbb{R}$ be a fixed frequency and $\phi_1, \phi_2 \in \mathbb{R}$ be phases. Using Euler's identity, show that

$$\cos(\omega t + \phi_1) + \cos(\omega t + \phi_2)$$

can be written as a single cosine of the same frequency in the form

$$A\cos(\omega t + \phi).$$

and derive expressions for A and ϕ.

Hint: You may find the identity $e^{i\phi_1} + e^{i\phi_2} = e^{i\left(\frac{\phi_1+\phi_2}{2}\right)}\left(e^{i\left(\frac{\phi_1-\phi_2}{2}\right)} + e^{-i\left(\frac{\phi_1-\phi_2}{2}\right)}\right)$ *useful.*

Solution: Using Euler's identity, we know we can rewrite

$$\cos(\theta) = \frac{1}{2}(e^{i\theta} + e^{-j\theta})$$

$$\begin{aligned}\cos(\omega t + \phi_1) + \cos(\omega t + \phi_2) &= \frac{1}{2}\left(e^{i(\omega t+\phi_1)} + e^{-i(\omega t+\phi_1)} + e^{i(\omega t+\phi_2)} + e^{-i(\omega t+\phi_2)}\right)\\ &= \frac{1}{2}\left(\left(e^{i\phi_1} + e^{i\phi_2}\right)e^{i\omega t} + \left(e^{-i\phi_1} + e^{-i\phi_2}\right)e^{-i\omega t}\right)\end{aligned}$$

We can simplify $e^{i\phi_1} + e^{i\phi_2}$ by factoring out the average, which is a common pattern.

$$\begin{aligned}e^{i\phi_1} + e^{i\phi_2} &= e^{i\left(\frac{\phi_1+\phi_2}{2}\right)}\left(e^{i\left(\frac{\phi_1-\phi_2}{2}\right)} + e^{-i\left(\frac{\phi_1-\phi_2}{2}\right)}\right)\\ &= e^{i\left(\frac{\phi_1+\phi_2}{2}\right)}\left(2\cos\left(\frac{\phi_1-\phi_2}{2}\right)\right)\end{aligned}$$

We can apply the same trick to simplify

$$e^{-i\phi_1} + e^{-i\phi_2} = 2\cos\left(\frac{\phi_1-\phi_2}{2}\right)e^{-i\left(\frac{\phi_1+\phi_2}{2}\right)}$$

Continuing to simplify using these new expressions,

$$\begin{aligned}\cos(\omega t + \phi_1) + \cos(\omega_t + \phi_2) &= \cos\left(\frac{\phi_1-\phi_2}{2}\right)\left(e^{i\omega t\left(\frac{\phi_1+\phi_2}{2}\right)} + e^{-i\omega t\left(\frac{\phi_1+\phi_2}{2}\right)}\right)\\ &= 2\cos\left(\frac{\phi_1-\phi_2}{2}\right)\cos\left(\omega t + \left(\frac{\phi_1+\phi_2}{2}\right)\right)\end{aligned}$$

Thus, we've demonstrated that the sum of two out-of-phase cosines with the same frequency can be written as a cosine of a single phase at that frequency, where

$$A = 2\cos\left(\frac{\phi_1-\phi_2}{2}\right), \quad \phi = \frac{\phi_1+\phi_2}{2}$$

(b) In fact, any linear combination of a set of sinewaves of frequency ω is always a sinewave of the same frequency, *even if* each sinewave has a distinct phase and amplitude. Show this fact. In particular, show that

$$\sum_{k=1}^{N} A_k \cos(\omega t + \phi_k) = A\cos(\omega t + \phi).$$

Recognize that

$$A\cos(\omega t + \phi) = \sum_{k=1}^{n} A_k \cos(\omega t + \phi_k)$$

can be obtained by equating the real parts of the two sides of

$$Ae^{i(\omega t+\phi)} = \sum_{k=1}^{n} A_k e^{i(\omega t+\phi_k)}.$$

Do away with $e^{i\omega t}$ from each side and simplify from there.

Solution: By the hint, write the complex sum

$$\sum_{k=1}^{N} A_k e^{i\phi_k} = Ae^{i\phi},$$

for suitable real A and ϕ (every complex number can be expressed in this polar form). Multiply both sides by $e^{i\omega t}$:

$$\sum_{k=1}^{N} A_k e^{i(\omega t+\phi_k)} = Ae^{i(\omega t+\phi)}.$$

Using Euler's formula $e^{i\theta} = \cos\theta + i\sin\theta$ and expanding yields

$$\sum_{k=1}^{N} A_k \big(\cos(\omega t + \phi_k) + i\sin(\omega t + \phi_k)\big) = A\big(\cos(\omega t + \phi) + i\sin(\omega t + \phi)\big).$$

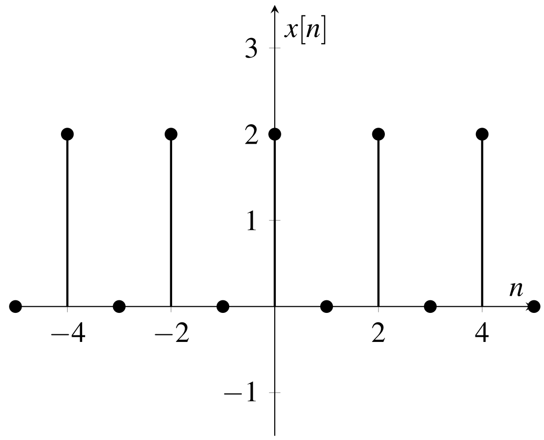
Equating real parts gives

$$\sum_{k=1}^{N} A_k \cos(\omega t + \phi_k) = A\cos(\omega t + \phi),$$

which is the desired identity.

5. DTFS Warmup (Adapted from EE 120 Discussion)

Consider the periodic DT signal $x[n]$ shown below:



(a) Determine the fundamental period p_x and fundamental frequency ω_0 of $x[n]$.

Solution:

The fundamental period of a periodic DT signal $x[n]$ is the minimum positive integer p such that

$$x[n+p] = x[n], \quad \forall n \in \mathbb{Z} \tag{1}$$

By inspection, we see $x[n]$ repeats every 2 time steps, so $p_x = 2$. To find the fundamental frequency, we use the formula $\omega = \frac{2\pi}{p_x}$ (radians per unit time).

$$\omega_0 = \frac{2\pi}{2} = \pi \tag{2}$$

(b) Determine an expression of the signal $x[n]$ as a linear combination of complex exponentials.

Solution:

We can observe that $x[n]$ can be expressed as

$$x[n] = (-1)^n + 1 = e^{i\pi n} + e^0$$

(c) Determine the complex exponential discrete-time Fourier series (DTFS) representation of $x[n]$. In particular, determine the values for the coefficients X_k in the DTFS expansion described by the analysis equation.

$$X_k = \frac{1}{p_x}\sum_{n=0}^{p_x-1} x[n]e^{-ik\omega_0 n} \tag{3}$$

using the p_x and ω_0 found in part (a).

Solution:

Because our period is $p_x = 2$, we can choose the period $\{0, 1\}$ to be our period of choice for analysis. Therefore, our DTFS coefficients are:

$$X_0 = \frac{1}{p_x}\sum_{n=0}^{p_x-1} x[n]e^{-i(0)\omega_0 n} = \frac{1}{2}(x[0]e^{-i(0)\pi(0)} + x[1]e^{-i(0)\pi(1)}) = \frac{1}{2}(2e^0 + 0e^0) = 1$$

$$X_1 = \frac{1}{p_x}\sum_{n=0}^{p_x-1} x[n]e^{-i(1)\omega_0 n} = \frac{1}{2}(x[0]e^{-i(1)\pi(0)} + x[1]e^{-i(1)\pi(1)}) = \frac{1}{2}(2e^0 + 0e^{-i\pi}) = 1$$

(d) As a sanity check, use the DTFS coefficients X_k to determine the equation for $x[n]$, and confirm that it matches your answer for part (b). You may find the synthesis equation helpful:

$$x[n] = \sum_{k=0}^{p_x-1} X_k e^{ik\omega_0 n} \tag{4}$$

Hint: Try splitting up the problem by solving for $x[n]$ when n is even and when it is odd.

Solution:

Let $\ell \in \mathbb{Z}$ be an integer value. It is the case that 2ℓ is always even and $2\ell+1$ is always odd. Using this, we can see that for all ℓ:

$$x[2\ell] = \sum_{k=0}^{p_x-1} X_k e^{ik\omega_0 n} = \sum_{k=0}^{1} (1) e^{ik\pi(2\ell)} = \sum_{k=0}^{1} e^{2\pi ik\ell} = \sum_{k=0}^{1} 1 = 1+1 = 2$$

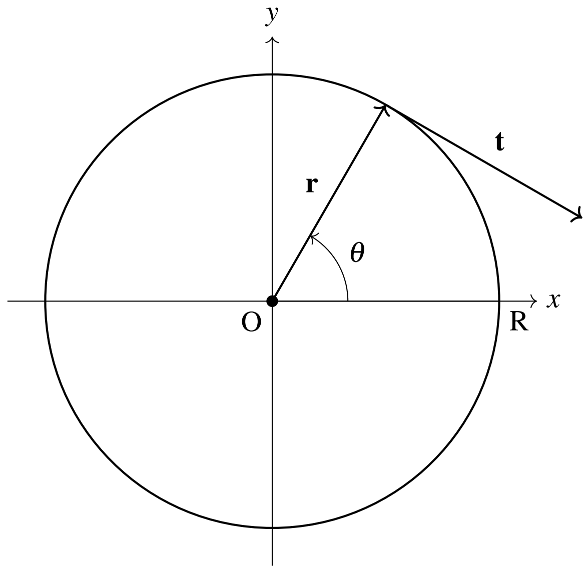
$$x[2\ell+1] = \sum_{k=0}^{p_x-1} X_k e^{ik\omega_0 n} = \sum_{k=0}^{1} (1) e^{ik\pi(2\ell+1)} = \sum_{k=0}^{1} e^{2\pi ik\ell+ik\pi} = \sum_{k=0}^{1} e^{2\pi ik\ell} e^{ik\pi} = (1)(1)+(1)(-1) = 0$$

We have demonstrated that under our assignments of the DTFS coefficients, $x[n] = 2$ for all even n and $x[n] = 0$ for all odd n, which is precisely the pattern we see in the signal.

Alternatively, we could solve specifically for the values of $x[0]$ and $x[1]$, and then argue that their values continue periodically across the whole signal.

6. Polar Orthogonality

Conside the polar coordinate system pictured in the diagram below:



Show mathematically that for any arbitrary radius R and angle θ, the vectors $\mathbf{r}$ and $\mathbf{t}$ ($\mathbf{t}$ points tangent to the circle starting at point $\mathbf{r}$) form an orthogonal basis for $\mathbb{R}^2$ in the Cartesian (xy) plane.

Hint: It may be useful to consider the parametric representation of a circle. How can we represent $\mathbf{r}$ *and* $\mathbf{t}$ *using this representation?*

Solution: Per the hint, we can recall the parametric equations for a circle are as follows: $x = R\cos\theta$, $y = R\sin\theta$

Therefore, in Cartesian coordinates, the vector $\mathbf{r} = \begin{bmatrix} R\cos\theta \\ R\sin\theta \end{bmatrix}$

As the tangent line, our vector $\mathbf{t}$ points in the direction of the slope of our curve at point $\mathbf{r}$. This slope, equal to $\frac{\partial y}{\partial x}$, can be computed as follows:

$\frac{\partial y}{\partial x} = \frac{\partial y/\partial\theta}{\partial x/\partial\theta}$
$\frac{\partial y}{\partial\theta} = \frac{\partial}{\partial\theta}(R\sin\theta) = R\cos\theta$
$\frac{\partial x}{\partial\theta} = \frac{\partial}{\partial\theta}(R\cos\theta) = -R\sin\theta$

$\frac{\partial y/\partial\theta}{\partial x/\partial\theta} = \frac{R\cos\theta}{-R\sin\theta} = -\frac{\cos\theta}{\sin\theta}$

For the purposes of orthogonality, the magnitude of $\mathbf{t}$ does not matter. As a result, we can capture its direction in the vector $\hat{\mathbf{t}} = \begin{bmatrix} 1 \\ -\frac{\cos\theta}{\sin\theta} \end{bmatrix}$

Recall that as the slope of the tangent line, $\frac{\partial y}{\partial x}$ describes the change in y for a unit change in x, hence the inclusion of 1 as the x-component of $\hat{\mathbf{t}}$ and $\frac{\partial y}{\partial x} = -\frac{\cos\theta}{\sin\theta}$ as the y-component of $\hat{\mathbf{t}}$.

Finally, we can take the inner product of $\mathbf{r}$ and $\hat{\mathbf{t}}$ in this Cartesian coordinate system:
$\langle \mathbf{r}, \hat{\mathbf{t}} \rangle = \mathbf{r}^T \hat{\mathbf{t}} = \begin{bmatrix} R\cos\theta & R\sin\theta \end{bmatrix} \begin{bmatrix} 1 \\ -\frac{\cos\theta}{\sin\theta} \end{bmatrix} = R\cos\theta \cdot 1 + R\sin\theta \cdot (-\frac{\cos\theta}{\sin\theta}) = R\cos\theta - R\cos\theta = 0$

Their inner product is 0, so $\mathbf{r}$ and $\hat{\mathbf{t}}$ must be orthogonal to one another. Because $\mathbf{t}$ is simply some scaling of $\hat{\mathbf{t}}$, it must also be orthogonal to $\mathbf{r}$.

Since $\mathbf{r}$ and $\mathbf{t}$ are two vectors orthogonal to one another in $\mathbb{R}^2$, they must be linearly independent of one another and must additionally span all of $\mathbb{R}^2$. As two orthogonal vectors that span all of $\mathbb{R}^2$, $\mathbf{r}$ and $\mathbf{t}$ serve as an orthogonal basis for $\mathbb{R}^2$.

7. Signal Analysis with Orthogonal Projections

In many applications, signals are mixtures of a few frequencies. In this problem, you will project a length-p signal onto a complex-exponential basis vectors and analyze the resultant coefficients.

This problem has an IPython component. Please attach a pdf printout of your IPython code to the HW in order to receive credit for problems that involve coding as stated in the submission format above.

(a) Write two Python functions:

(i) `complex_inner_product(u, v)`: takes two complex vectors $\mathbf{u}, \mathbf{v} \in \mathbb{C}^p$ and returns $\langle \mathbf{u}, \mathbf{v} \rangle$.

(ii) `generate_basis_vector(k, p)`: returns the DTFS basis vector $\boldsymbol{\psi}_k$ of length p with entries $(\boldsymbol{\psi}_k)[n] = e^{i2\pi nk/p}$.

(iii) Using your functions, with $p = 64$, numerically evaluate $\langle \boldsymbol{\psi}_3, \boldsymbol{\psi}_{10} \rangle$ and report its magnitude.

Solution: Your functions should implement the definitions verbatim: $\sum_{n=0}^{p-1} u[n]v[n]^*$ for the inner product, and compute $e^{i2\pi nk/p}$ for each component of $\boldsymbol{\psi}_k$. Orthogonality implies $\langle \boldsymbol{\psi}_3, \boldsymbol{\psi}_{10} \rangle = 0$ in exact arithmetic, but numerically, the magnitude might be slightly off 0 due to floating point arithmetic. See the solution IPython notebook for an example of a correct implementation.

(b) Let $p = 64$, $f_1 = 2$ Hz, $f_2 = 8$ Hz, $A_1 = 2.0$, $A_2 = 0.5$. Define $t_n = n/p$ and $s : \mathbb{R} \to \mathbb{R}$ as

$$s(t) = A_1 \cos(2\pi f_1 t) + A_2 \cos(2\pi f_2 t).$$

Write a Python function `generate_signal(p, frequencies, amplitudes)` that returns $\mathbf{s} \in \mathbb{R}^{64}$ with entries $s[n] = s(t_n)$ for these parameters. Additionally report the first four samples $s[0:4]$. Note that instead of s being a discrete-time signal, it is a continuous-time signal for which we take discrete samples, which gives us the vector $\mathbf{s}$.

Solution: Compute $t_n = n/64$ and evaluate the cosine sum at each t_n to obtain a length-64 real vector. See the solution IPython notebook for an example of a correct implementation.

(c) The contribution of frequency index k in $\mathbf{s}$ is given by the projection coefficient

$$c_k = \mathrm{proj}_{\boldsymbol{\psi}_k}(\mathbf{s}) = \frac{\langle \mathbf{s}, \boldsymbol{\psi}_k \rangle}{\langle \boldsymbol{\psi}_k, \boldsymbol{\psi}_k \rangle} = \frac{1}{p} \langle \mathbf{s}, \boldsymbol{\psi}_k \rangle.$$

Write a Python function `get_projection_coefficient(signal, k, p)` that computes c_k using your tools from part (a).

Solution: Implement $c_k = \frac{1}{p} \sum_{n=0}^{p-1} s[n](\boldsymbol{\psi}_k)[n]^*$. See the solution IPython notebook for an example of a correct implementation.

(d) For $p = 64$, $k = 0, 1, \ldots, 31$, compute c_k and plot $|c_k|$ versus k, with labeled axes. Are there peaks at $k = 2$ and $k = 8$? How do the magnitudes relate to A_1 and A_2?

Solution: With $p = 64$ and $f_1 = 2$, $f_2 = 8$, peaks should occur at $k = 2$ and $k = 8$. Magnitudes reflect A_1 and A_2 but halved. See the solution IPython notebook for an example of a correct implementation and plot.

(e) Repeat parts (b) through (d) with the following frequencies and amplitudes: $f_1 = 24$, $f_2 = 26$, $A_1 = 1$, $A_2 = 1$. We will refer to this signal as s_2. Afterwards, generate a plot of the discrete-time signal $s_2[n]$. Describe the signal in words. Does the signal match what you'd expect from the DTFS coefficients?

Solution: With $p = 64$ and $f_1 = 24$, $f_2 = 26$, peaks should occur at $k = 24$ and $k = 26$. Magnitudes reflect A_1 and A_2 but halved, as before. When plotting the discrete-time signal, we see that while it is a sum of two cosines of different frequency, the resultant signal is a 25 Hz signal modulated by a 2 Hz signal. Neither of these frequencies are present in the DTFS coefficients. See the solution IPython notebook for an example of a correct implementation and plot.

8. Homework Process and Study Group

Whom did you work with on this homework? List names and student ID's. (In case you met people at homework party or in office hours, you can also just describe the group.) How did you work on this homework? If you worked in your study group, explain what role each student played for the meetings this week.

Solution:

I first worked by myself for 2 hours, but got stuck on problem 5. Then I met with my study group.

XYZ played the role of facilitator ... etc. We were still stuck on problem 5 so we went to office hours to talk about the problem.

Then I went to homework party for a few hours, where I finished the homework.